

ST6101: Advanced Statistical Theory (I)

Lecture 4: Sufficiency and Completeness

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*Adapted from Dr. Dongming Huang's slides.

Recap: Sufficiency and factorization

Let X be a sample from an unknown population $P \in \mathcal{P}$, where $\mathcal{P}$ is a family of populations.

- A statistic $T(X)$ is said to be **sufficient** for $P \in \mathcal{P}$ iff the conditional distribution of X given T is known (does not depend on P)
- **Factorization Theorem:** Further suppose that $\mathcal{P}$ is a family of probability measures on $(\mathbb{R}^n, \mathcal{B}^n)$ dominated by a σ -finite measure ν .

Then a statistic $T(X)$ is sufficient for $P \in \mathcal{P}$ **if and only if** there are nonnegative Borel functions $h(x)$ on $(\mathbb{R}^n, \mathcal{B}^n)$ and $g_P(t)$ on the range of T such that

$$\frac{dP}{d\nu}(x) = g_P(T(x))h(x).$$

Factorization Theorem

Theorem

Suppose that X is a sample from $P \in \mathcal{P}$ and $\mathcal{P}$ is a family of probability measures on $(\mathbb{R}^n, \mathcal{B}^n)$ dominated by a σ -finite measure ν . Then $T(X)$ is sufficient for $P \in \mathcal{P}$ **if and only if** there are nonnegative Borel functions

- $h(x)$ (which does not depend on P) on $(\mathbb{R}^n, \mathcal{B}^n)$, and
- $g_P(t)$ (which depends on P) on the range of T

such that

$$\frac{dP}{d\nu}(x) = g_P(T(x))h(x).$$

Assume T is sufficient, and want to show $\frac{dP}{d\nu}(x) = g_P(T(x))h(x)$

- Note that $P(A) = \int P(A | T) dP$ for any $A \in \mathcal{B}^n$
- Suppose we can find a probability measure Q on $(\mathcal{R}^n, \mathcal{B}^n)$ that **dominates $\mathcal{P}$ and for all $P \in \mathcal{P}$, $P(A | T) = E_Q(I_A | T)$, Q -a.s.**
- On the measurable space $(\mathcal{R}^n, \sigma(T))$, let $\frac{dP}{dQ}(x)$ be the Radon-Nikodym derivative of P w.r.t. Q and denoted by $g_P(T(x))$.

$$\begin{aligned} P(A) &= \int P(A | T) dP &&= \int E_Q(I_A | T) g_P(T(x)) dQ \\ &&&= \int E_Q(I_A g_P(T(x)) | T) dQ \\ &&&= \int I_A g_P(T(x)) dQ \\ &&&= \int_A g_P(T(x)) \frac{dQ}{d\nu}(x) d\nu(x). \end{aligned}$$

- So $\frac{dP}{d\nu}(x) = g_P(T(x))h(x)$ holds with $h(x) = \frac{dQ}{d\nu}(x)$.

Finding Q

We need two lemmas.

Lemma (S, Exercise 1.35)

Let $\{c_i\}$ be a sequence of positive numbers satisfying $\sum_{i=1}^{\infty} c_i = 1$ and let $\{P_i\}$ be a sequence of probability measures on a common measurable space. Define $Q = \sum_{i=1}^{\infty} c_i P_i$.

- 1 Q is a probability measure;
- 2 Let ν be a σ -finite measure. Then $P_i \ll \nu$ for all i if and only if $Q \ll \nu$. When $Q \ll \nu$,

$$\frac{dQ}{d\nu} = \sum_{i=1}^{\infty} c_i \frac{dP_i}{d\nu}.$$

Lemma (2.1)

If a family $\mathcal{P}$ is dominated by a σ -finite measure ν , then $\mathcal{P}$ is dominated by a probability measure $Q = \sum_{i=1}^{\infty} c_i P_i$ where $P_i \in \mathcal{P}$ and $c_i \geq 0$.

Showing $P(A | T) = E_Q(I_A | T)$

Let Q be the one in Lemma 2.1. For any $B \in \sigma(T)$, by Lemma S,

$$\begin{aligned} Q(A \cap B) &= \sum_{j=1}^{\infty} c_j P_j(A \cap B) = \sum_{j=1}^{\infty} c_j \int_B P_j(A | T) dP_j \\ (\because P(A | T) \text{ doesn't depend on } P \in \mathcal{P}) &= \sum_{j=1}^{\infty} c_j \int_B P(A | T) dP_j \\ (\because \text{calculus of R-N derivatives}) &= \sum_{j=1}^{\infty} c_j \int_B P(A | T) \frac{dP_j}{dQ} dQ \\ (\because \text{Fubini's theorem}) &= \int_B \sum_{j=1}^{\infty} c_j P(A | T) \frac{dP_j}{dQ} dQ \\ (\because \sum_j c_j P_j = Q) &= \int_B P(A | T) dQ, \end{aligned}$$

Hence, $P(A | T) = E_Q(I_A | T)$ Q -a.s.

Assume $\frac{dP}{d\nu}(x) = g_P(T(x))h(x)$ holds, and want to show T is sufficient.

- Let Q be the probability measure in Lemma 2.1
- Want to show for all $A \in \sigma(X)$ and $P \in \mathcal{P}$,

$$P(A \mid T) = E_Q(I_A \mid T) \quad P\text{-a.s.}, \quad (1)$$

because $E_Q(I_A \mid T)$ does not vary with $P \in \mathcal{P}$.

We first show that dP/dQ is a Borel function of T :

- By Chain rule, $\frac{dP}{d\nu} = \frac{dP}{dQ} \frac{dQ}{d\nu}$ ν -a.e.
- Hence

$$\frac{dP}{dQ} = \frac{dP}{d\nu} \bigg/ \frac{dQ}{d\nu} = \frac{dP}{d\nu} \bigg/ \sum_{i=1}^{\infty} c_i \frac{dP_i}{d\nu} = g_P(T) \bigg/ \sum_{i=1}^{\infty} c_i g_{P_i}(T), \quad (2)$$

Q -a.s., where the second equality follows from Lemma S.

Showing $P(A | T) = E_Q(I_A | T)$

For any $B \in \sigma(T)$, we calculate $P(A \cap B)$ as follows

$$\begin{aligned}\int_B I_A dP &= \int_B I_A \frac{dP}{dQ} dQ \\ (\because \text{Def. of conditional expectation}) &= \int_B E_Q \left(I_A \frac{dP}{dQ} \mid T \right) dQ \\ (\because dP/dQ \text{ is a Borel function of } T) &= \int_B E_Q(I_A \mid T) \frac{dP}{dQ} dQ \\ &= \int_B E_Q(I_A \mid T) dP.\end{aligned}$$

So $P(A | T) = E_Q(I_A | T)$, P -a.s.

Exercise

Let X be a sample from $P \in \mathcal{P}$, where $\mathcal{P}$ is a family of distributions on $(\mathcal{R}^n, \mathcal{B}^n)$ dominated by a σ -finite measure ν .

Show that if $T(X)$ is a sufficient statistic for $P \in \mathcal{P}$ and $T = \psi(S)$, where ψ is measurable and $S(X)$ is another statistic, then $S(X)$ is also sufficient for $P \in \mathcal{P}$

Solution.

- By the factorization theorem,

$$\frac{dP}{d\nu}(x) = g_P(T(x))h(x)$$

where h is a Borel function of x and $g_P(t)$ is a Borel function of t

- If $T = \psi(S)$, then

$$\frac{dP}{d\nu}(x) = g_P(\psi(S(x)))h(x)$$

and, by the factorization theorem again, $S(X)$ is sufficient for $P \in \mathcal{P}$

- If $T(X) = \psi(S(X))$ and T is sufficient, then S is also sufficient
 - Knowledge of S implies knowledge of T
 - T provides a greater reduction of the data than S (unless ψ is one-to-one)
- Can we find the sufficient statistic that provides the greatest reduction?

Definition (Minimal sufficiency)

Let T be a sufficient statistic for $P \in \mathcal{P}$. T is called a minimal sufficient statistic if and only if, for any other statistic S sufficient for $P \in \mathcal{P}$, there is a measurable function ψ such that $T = \psi(S)$ $\mathcal{P}$ -a.s.

- Notation: If a statement holds except for outcomes in an event A satisfying $P(A) = 0$ for all $P \in \mathcal{P}$, then we say that the statement holds $\mathcal{P}$ -a.s.

- Minimal sufficient statistics are **unique** (almost surely) in the following sense
 - If both T and S are minimal sufficient statistics (for a family $\mathcal{P}$), then there is a one-to-one measurable function ψ such that $T = \psi(S)$ $\mathcal{P}$ -a.s.
- Minimal sufficient statistics **exist** under weak conditions
 - e.g., $\mathcal{P}$ contains distributions on $\mathbb{R}^n$ dominated by a σ -finite measure (Bahadur, 1957)

Theorems for finding minimal sufficient statistics

We have several useful tools for checking minimal sufficiency.
In the following slides, $\mathcal{P}$ is a family of distributions on $\mathbb{R}^n$.

Theorem (A)

Suppose $\mathcal{P}_0 \subset \mathcal{P}$ and $\mathcal{P}_0$ -a.s. implies $\mathcal{P}$ -a.s.

If T is sufficient for $P \in \mathcal{P}$ and minimal sufficient for $P \in \mathcal{P}_0$, then T is minimal sufficient for $P \in \mathcal{P}$.

Proof:

- If S is sufficient for $\mathcal{P}$, then it is also sufficient for $\mathcal{P}_0$.
- Thus, $T = \psi(S)$ $\mathcal{P}_0$ -a.s. for a measurable function ψ
- Then $T = \psi(S)$ $\mathcal{P}$ -a.s. since $\mathcal{P}_0$ -a.s. implies $\mathcal{P}$ -a.s by assumption.

Theorem (B)

Suppose that $\mathcal{P}$ contains p.d.f.'s $f_0, f_1, \dots$ w.r.t. a σ -finite measure.

- ① Define $f_\infty(x) = \sum_{i=0}^{\infty} c_i f_i(x)$, where $c_i > 0$ and $\sum_{i=0}^{\infty} c_i = 1$. Let $T_i(x) = f_i(x)/f_\infty(x)$ when $f_\infty(x) > 0$. Then $T(X) = (T_0(X), T_1(X), \dots)$ is minimal sufficient for $\mathcal{P}$.
- ② If $\{x : f_i(x) > 0\} \subset \{x : f_0(x) > 0\}$ for all i , then $T(X) = (f_1(x)/f_0(x), f_2(x)/f_0(x), \dots)$ is minimal sufficient for $\mathcal{P}$.

Proof of (1):

- The construction of f_∞ assures that $f_\infty(X) > 0$ $\mathcal{P}$ -a.s.
- Let $g_i(T) = T_i$ for all i . Then $f_i(x) = T_i(x)f_\infty(x) = g_i(T(x))f_\infty(x)$. By Factorization theorem, T is sufficient for $\mathcal{P}$.
- Suppose $S(X)$ is another sufficient statistic. By Factorization theorem, $f_i(x) = \tilde{g}_i(S(x))h(x)$ for all i and some $\tilde{g}_i$ and h .
- Then $T_i(x) = \tilde{g}_i(S(x))/\sum_{j=0}^{\infty} c_j \tilde{g}_j(S(x))$ wherever $f_\infty(x) > 0$.
- By definition, T is minimal sufficient for $\mathcal{P}$.

The argument for (2) is similar.

Theorem (C)

Suppose that $\mathcal{P}$ contains p.d.f.'s f_P w.r.t. a σ -finite measure ν . Suppose $T(X)$ a sufficient statistic that satisfies the following property:

If there is a measurable function ϕ such that for any possible values x and y of X ,

$$f_P(x) = f_P(y)\phi(x, y), \quad \forall P \in \mathcal{P}, \quad (3)$$

then $T(x) = T(y)$.

$T(X)$ is minimal sufficient for $\mathcal{P}$.

- Eq (3) can be understood as (inaccurately):
the density ratio $f_P(x)/f_P(y)$ does not depend on $P \in \mathcal{P}$.
- The support of the densities may vary, which may make the “ratio” undefined; however, Eq (3) can still be analyzed.

Idea behind Theorem (C)

See the textbook (Theorem 2.3.iii). Here is the basic idea.

- Suppose S is sufficient. Need to show T is a function of S .
- By factorization theorem, we have $f_P(x) = g_P(S(x))h(x)$, ν -a.e. , for any $P \in \mathcal{P}$.
- Let $A = \{x : h(x) > 0\}$. Then $P(A^c) = 0$ for any $P \in \mathcal{P}$.
- For any two values x and y in A s.t. $S(x) = S(y)$, then $\forall P \in \mathcal{P}$,

$$f_P(x) = g_P(S(x))h(x) = g_P(S(y))h(y) \times \frac{h(x)}{h(y)} = f_P(y) \times \frac{h(x)}{h(y)}.$$

Condition (3) implies $T(x) = T(y)$. So $T(x) = \psi(S(x))$, $\mathcal{P}$ -a.s.

Example: Normal families

- Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, both μ and σ^2 unknown.
- Let $\mathbf{x}$ and $\mathbf{y}$ denote two sample points
- Let $(\bar{x}, s_{\mathbf{x}}^2)$ and $(\bar{y}, s_{\mathbf{y}}^2)$ be the sample means and variances corresponding to the $\mathbf{x}$ and $\mathbf{y}$ samples, respectively
- The ratio of densities is

$$\begin{aligned}\frac{f(\mathbf{x} \mid \mu, \sigma^2)}{f(\mathbf{y} \mid \mu, \sigma^2)} &= \frac{(2\pi\sigma^2)^{-n/2} \exp\left(-\left[n(\bar{x} - \mu)^2 + (n-1)s_{\mathbf{x}}^2\right] / (2\sigma^2)\right)}{(2\pi\sigma^2)^{-n/2} \exp\left(-\left[n(\bar{y} - \mu)^2 + (n-1)s_{\mathbf{y}}^2\right] / (2\sigma^2)\right)} \\ &= \exp\left(\left((2\sigma^2)^{-1} \left[-n(\bar{x}^2 - \bar{y}^2) + 2n\mu(\bar{x} - \bar{y}) - (n-1)(s_{\mathbf{x}}^2 - s_{\mathbf{y}}^2)\right]\right)\right)\end{aligned}$$

- This ratio will be constant as a function of μ and σ^2 if and only if $\bar{x} = \bar{y}$ and $s_{\mathbf{x}}^2 = s_{\mathbf{y}}^2$
- Thus, by Theorem C, $(\bar{X}, S^2)$ is a minimal sufficient statistic for (μ, σ^2)

Example: Exponential family

Let $\mathcal{P} = \{f_\theta : \theta \in \Theta\}$ be an exponential family with p.d.f.'s

$$f_\theta(x) = \exp\{[\eta(\theta)]^\top T(x) - \xi(\theta)\}h(x),$$

where η and T are p -dimensional.

- By Factorization Theorem, $T(X)$ is sufficient for $\theta \in \Theta$
- T is also minimal sufficient, if the following holds:
There exists $\Theta_0 = \{\theta_0, \theta_1, \dots, \theta_p\} \subset \Theta$ such that the vectors $\eta_i = \eta(\theta_i) - \eta(\theta_0)$, $i = 1, \dots, p$, are linearly independent in $\mathcal{R}^p$,
 - Algebraically, check whether the determinant of the matrix $[\eta_1, \dots, \eta_p]$ is nonzero.
 - Geometrically, it means the natural parameter space $\Xi = \{\eta(\theta) : \theta \in \Theta\}$ contains $(p+1)$ points that do not lie on the same hyperplane.
 - This is true if the corresponding natural exponential family is of full rank.
- Two methods to prove
 - Method 1: combining Theorems A and B
 - Method 2: using Theorem C

Example (Cont.): Method 1

Let $\mathcal{P} = \{f_\theta : \theta \in \Theta\}$ be a p -dimensional exponential family with p.d.f.'s

$$f_\theta(x) = \exp\{[\eta(\theta)]^\top T(x) - \xi(\theta)\}h(x).$$

If there exists $\Theta_0 = \{\theta_0, \theta_1, \dots, \theta_p\} \subset \Theta$ such that the vectors $\eta_i = \eta(\theta_i) - \eta(\theta_0)$, $i = 1, \dots, p$, are linearly independent in $\mathcal{R}^p$, then T is minimal sufficient.

- Let $\mathcal{P}_0 = \{f_\theta : \theta \in \Theta_0\}$
- Note that the set $\{x : f_\theta(x) > 0\}$ does not depend on θ
- It follows from Theorem B with f_{θ_0} that

$$S(X) = \left(\exp\{\eta_1^\top T(x) - \xi_1\}, \dots, \exp\{\eta_p^\top T(x) - \xi_p\} \right)$$

is minimal sufficient for $\theta \in \Theta_0$

- Since η_i 's are linearly independent, there is a one-to-one measurable function ψ such that $T(X) = \psi(S(X))$ a.s. $\mathcal{P}_0$
- Hence, T is minimal sufficient for $\theta \in \Theta_0$
- It is easy to see that $\mathcal{P}_0$ -a.s. implies $\mathcal{P}$ -a.s.
- By Theorem A, T is minimal sufficient for $\theta \in \Theta$.

Example (Cont.): Method 2

Let $\mathcal{P} = \{f_\theta : \theta \in \Theta\}$ be a p -dimensional exponential family with p.d.f.'s

$$f_\theta(x) = \exp\{[\eta(\theta)]^\top T(x) - \xi(\theta)\}h(x).$$

If there exists $\Theta_0 = \{\theta_0, \theta_1, \dots, \theta_p\} \subset \Theta$ such that the vectors $\eta_i = \eta(\theta_i) - \eta(\theta_0)$, $i = 1, \dots, p$, are linearly independent in $\mathcal{R}^p$, then T is minimal sufficient.

- Suppose $x, y \in \{x : h(x) > 0\}$ such that

$$f_\theta(x) = f_\theta(y)\phi(x, y), \forall \theta \in \Theta,$$

where $\phi(x, y)$ does not depend on θ .

- By restricting on Θ_0 , we have

$$\exp\{[\eta(\theta_i) - \eta(\theta_0)]^\top [T(x) - T(y)]\} = 1, \forall i = 1, \dots, p.$$

- Since $\{\eta_i\}_{i=1}^p$ are linearly independent, we have $T(x) = T(y)$.
- Since T is sufficient, by Theorem C, T is minimal sufficient.

- A minimal sufficient statistic might not be “simplest sufficient statistic”
 - May still contain redundant information
 - e.g.: if T is minimal sufficient, then so is $S = (T + 1, e^T)$
- Need a notion that describes whether a statistic contains “redundant information”
- Before that, we introduce a notion for “having no relevant information”

Ancillary statistics

- A statistic $V(X)$ is said to be **ancillary** for $\mathcal{P}$ if its distribution does not depend on the population $P \in \mathcal{P}$;
 $V(X)$ is said to be **first-order ancillary** for $\mathcal{P}$ if $E_P[V(X)]$ does not depend on $P \in \mathcal{P}$
 - e.g.: trivial ancillary statistic: $V(X) = c$
- If $V(X)$ is a nontrivial ancillary statistic, then $\sigma(V(X))$ is a nontrivial σ -field that does not contain any information about P
 - Being ancillary means that for any $A \in \sigma(V(X))$, $P(A)$ is constant across all $P \in \mathcal{P}$
 - We cannot learn anything about P if we only look at events in $\sigma(V(X))$
- Similarly, for a statistic $S(X)$, if $V(S(X))$ is a nontrivial ancillary statistic, then $\sigma(S(X))$ contains a nontrivial σ -field $\sigma(V(S(X)))$ that does not contain any information about P
 - The “data” $S(X)$ may be further reduced
- Intuition: A sufficient statistic T
is good in reducing data *if no non-constant function of T is ancillary or even first-order ancillary*

Definition

- A statistic $T(X)$ is said to be **complete** for $P \in \mathcal{P}$ if and only if, for any Borel function f , $E_P f(T) = 0$ for all $P \in \mathcal{P}$ implies that $f(T) = 0$ $\mathcal{P}$ -a.s.
- T is said to be **boundedly complete** if and only if the previous statement holds for any bounded Borel functions f .
- Intuition: A complete statistic contains “completely” useful information about P ; no redundancy
- Clearly, a complete statistic is boundedly complete
- If T is complete and $S = \psi(T)$ for a measurable function ψ , then S is also complete
 - Similar result holds for bounded completeness
- A complete sufficient statistic is effective in reducing the data, so we expect that a complete sufficient statistic is always minimal sufficient

Completeness + Sufficiency $\Rightarrow$ Minimal Sufficiency

Proposition

Suppose X is a sample from unknown $P \in \mathcal{P}$, and suppose a minimal sufficient statistic exists.

If a statistic U is sufficient and boundedly complete, then U is minimal sufficient.

We first assume U is one-dimensional

- Suppose T is minimal sufficient. We have $T = \varphi(U)$
- We want to show that $U = \psi(T)$ for some ψ
- Let $h : \mathbb{R} \mapsto \mathbb{R}$ be a bounded 1-1 continuous function
- Note that $E_P[h(U) \mid T]$ does not depend on P and can be written as $\eta(T)$. Easy to see that η is also bounded
- $E_P[h(U) - \eta(T)] = E_P[h(U) - E_P[h(U) \mid T]] = 0$
- So $E_P[h(U) - \eta(\varphi(U))] = 0$ for all $P \in \mathcal{P}$. Since U is boundedly complete, we have $h(U) - \eta(T) = 0$ $\mathcal{P}$ -a.s.
- Therefore $U = h^{-1}(\eta(T))$ $\mathcal{P}$ -a.s.

If U is p -dimensional, we use the same argument with two modifications:

- Use the notation $h(U)$ to denote the vector $(h(U_1), h(U_2), \dots, h(U_p))$, and $\eta(T)$ to denote the vector $(E_P[h(U_1) \mid T], \dots, E_P[h(U_p) \mid T])$
- For each $i = 1, \dots, p$, it holds that $E_P[(h(U))_i - (\eta(\varphi(U)))_i] = 0$.
- By the completeness of U , we conclude that $h(U) = \eta(T)$ a.s.

Remark. A minimal sufficient statistic is **not** necessarily complete
Example:

- $P \in \{\text{Unif}(\theta, \theta + 1) : \theta \in \mathbb{R}\}$
- $T(X) = (X_{(1)}, X_{(n)})$ is minimal sufficient
- But $T_2 - T_1 = X_{(n)} - X_{(1)}$ does not depend on P , so T is not complete.

Complete Sufficient Statistics in Exponential Families

Proposition (A)

If $\mathcal{P}$ is a natural exponential family of full rank with p.d.f.'s given by

$$p_{\eta}(x) = \exp\{\eta^{\top}T(x) - \zeta(\eta)\}h(x),$$

then $T(X)$ is complete and sufficient for $\eta \in \Xi$.

Remark. This result provides another way to show T is minimal sufficient.

Proof: We have shown that T is sufficient. Now show T is complete.

Suppose f is a function such that $E_{\eta}[f(T)] = 0$ for all $\eta \in \Xi$, i.e.,

$$\int f(t) \exp\{\eta^{\top}t - \zeta(\eta)\}d\lambda = 0 \quad \text{for all } \eta \in \Xi,$$

where λ is a measure on $(\mathcal{R}^p, \mathcal{B}^p)$ (Theorem 2.1 in the textbook)

Exponential Families (Cont.)

- Let η_0 be an interior point of Ξ
- Then we can find a neighborhood of η_0 ,
 $N(\eta_0) = \{\eta \in \mathcal{R}^p : \|\eta - \eta_0\| < \epsilon\}$ for some $\epsilon > 0$, such that

$$\int f_+(t) e^{\eta^\top t} d\lambda = \int f_-(t) e^{\eta^\top t} d\lambda, \quad \forall \eta \in N(\eta_0). \quad (4)$$

- In particular,

$$\int f_+(t) e^{\eta_0^\top t} d\lambda = \int f_-(t) e^{\eta_0^\top t} d\lambda = c.$$

- If $c = 0$, then $f = 0$ a.e. λ
- If $c > 0$, then $c^{-1}f_+(t)e^{\eta_0^\top t}$ and $c^{-1}f_-(t)e^{\eta_0^\top t}$ are p.d.f.'s w.r.t. λ
 - Eq. (4) implies that their m.g.f.'s are the same in a neighborhood of 0
 - By the uniqueness of the m.g.f., we conclude the two p.d.f.'s are the same λ -a.e., which implies $f = f_+ - f_- = 0$ λ -a.e.
- Hence T is complete.

Example: normal families

Suppose that $X_1, \dots, X_n$ are i.i.d. random variables having the $N(\mu, \sigma^2)$ distribution, $\mu \in \mathbb{R}$, $\sigma > 0$ are **unknown**

- It is easy to check that the joint p.d.f. is

$$(2\pi)^{-n/2} \exp \{ \eta_1 T_1 + \eta_2 T_2 - n\zeta(n) \},$$

where

- $T_1 = \sum_{i=1}^n X_i$
- $T_2 = -\sum_{i=1}^n X_i^2$
- $\eta = (\eta_1, \eta_2) = (\mu/\sigma^2, 1/(2\sigma^2))$
- This is a natural exponential family of full rank: $\Xi = \mathbb{R} \times (0, \infty)$ is an open set of $\mathbb{R}^2$
- So $T(X) = (T_1(X), T_2(X))$ is complete and sufficient for η
- There is a one-to-one correspondence between η and $\theta = (\mu, \sigma^2)$
 - T is also complete and sufficient for θ
- There is a one-to-one correspondence between $(\bar{X}, S^2)$ and (T_1, T_2)
 - $(\bar{X}, S^2)$ is complete and sufficient for θ

Example: Uniform Family with Varying Right Ends

Let $X_1, \dots, X_n \sim P_\theta = U(0, \theta)$ be i.i.d. for $\theta > 0$. We will show that the largest order statistic, $X_{(n)}$, is complete and sufficient for θ

- The sufficiency follows from Factorization theorem:
the joint p.d.f. is $\theta^{-n} I_{(0,\theta)}(x_{(n)}) I_{(0,\infty)}(x_{(1)})$
- The CDF of $X_{(n)}$ is

$$\begin{aligned} &= P_\theta(X_{(n)} \leq x) = P_\theta(X_1 \leq x, \dots, X_n \leq x) \\ &= \prod_{i=1}^n P_\theta(X_i \leq x) = \frac{x^n}{\theta^n} I_{(0,\theta)}(x). \end{aligned}$$

- The p.d.f. of $X_{(n)}$ is then

$$\frac{nx^{n-1}}{\theta^n} I_{(0,\theta)}(x).$$

Example (Cont.)

- Let g be a Borel function on $[0, \infty)$ s.t. $E_\theta[g(X_{(n)})] = 0$ for all $\theta > 0$.
Then

$$\int_0^\theta g(x)x^{n-1}dx = 0$$

for all $\theta > 0$

- Differentiate** the above w.r.t. θ

$$g(\theta)\theta^{n-1} = 0, \quad a.e.$$

- Thus, $g(\theta) = 0$ a.e. on $(0, \infty)$ and $g(X_{(n)}) = 0$ $\mathcal{P}$ -a.s.
- By definition, $X_{(n)}$ is complete for θ

Basu's Theorem

The relationship between an ancillary statistic and a complete and sufficient statistic is characterized in the following result

Theorem (Basu)

Let V and T be two statistics of X from a population $P \in \mathcal{P}$. If V is ancillary and T is boundedly complete and sufficient for $P \in \mathcal{P}$, then V and T are independent w.r.t. any $P \in \mathcal{P}$.

- Intuition: $\sigma(V)$ does not contain information about P , while $\sigma(T)$ carries non-redundant and sufficient information about P . This suggests that V and T are independent
- Basu's theorem is useful in proving the independence of two statistics.

Proof:

- Let A be a Borel set on the range of T and B be a Borel set on the range of V . Need to show

$$P(T \in A, V \in B) = P(T \in A)P(V \in B), \quad \forall P \in \mathcal{P}$$

- Since V is ancillary, $P(V \in B)$ is a constant c_B
- Since T is sufficient, $E_P[I_B(V)|T]$ can be written as $h_B(T)$
- By tower property, for any $P \in \mathcal{P}$, it holds that
$$E_P[h_B(T) - c_B] = E_P\{E_P[I_B(V)|T] - P(V \in B)\} = 0$$
- By the bounded completeness of T , $h_B(T) - c_B = 0$ $\mathcal{P}$ -a.s.
- By the properties of conditional expectation, for all $P \in \mathcal{P}$,

$$\begin{aligned} P(T \in A, V \in B) &= E_P\{E_P[I_A(T)I_B(V)|T]\} \\ &= E_P\{I_A(T)E_P[I_B(V)|T]\} \\ &= E_P\{I_A(T)P(V \in B)\} \\ &= P(T \in A)P(V \in B), \end{aligned}$$

where the 3rd equation is due to $h_B(T) - c_B = 0$, P -a.s.

Example: $\bar{X}$ and S^2 of a normal sample

If $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, then the sample mean $\bar{X}$ and sample variance S^2 are independent.

*This result does not rely on any model, but we can **postulate a model** and prove it using Basu's theorem w.r.t. the model.*

- Consider the family $\{N(\mu, \sigma^2) : \mu \in \mathbb{R}\}$ with σ^2 known
- Natural parameter: $\eta = \mu/\sigma^2$
- Easy to show it is an exponential family of full rank
- Then $\bar{X}$ is complete and sufficient for μ according to Proposition A
- Rewrite $S^2 = (n-1)^{-1} \sum_{i=1}^n (Z_i - \bar{Z})^2$, where $Z_i = X_i - \mu \sim N(0, \sigma^2)$ and $\bar{Z}$ is the sample mean of Z_i 's.
 S^2 is ancillary w.r.t. μ , since its distribution does not depend on μ
- By Basu's theorem, $\bar{X}$ and S^2 are independent w.r.t. $N(\mu, \sigma^2)$ for any $\mu \in \mathbb{R}$
- Since σ^2 is arbitrary, $\bar{X}$ and S^2 are independent w.r.t. $N(\mu, \sigma^2)$ for any $\mu \in \mathbb{R}$ and $\sigma^2 > 0$

Example (Cont.)

Using the independence of $\bar{X}$ and S^2 , we can show that $(n-1)S^2/\sigma^2$ has the chi-square distribution χ_{n-1}^2

- Note that

$$n \left(\frac{\bar{X} - \mu}{\sigma} \right)^2 + \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2,$$

so the m.g.f. of the RHS equals to the product of the m.g.f.'s of $n(\bar{X} - \mu)^2/\sigma^2$ and $(n-1)S^2/\sigma^2$

- $n(\bar{X} - \mu)^2/\sigma^2$ has the chi-square distribution χ_1^2 with the m.g.f. $(1-2t)^{-1/2}, t < 1/2$
- $\sum_{i=1}^n (X_i - \mu)^2/\sigma^2$ has the chi-square distribution χ_n^2 with the m.g.f. $(1-2t)^{-n/2}, t < 1/2$
- By the independence of $\bar{X}$ and S^2 , the m.g.f. of $(n-1)S^2/\sigma^2$ is

$$(1-2t)^{-n/2} / (1-2t)^{-1/2} = (1-2t)^{-(n-1)/2}$$

for $t < 1/2$

- This is the m.g.f. of the chi-square distribution χ_{n-1}^2